A

Automorphic Hecke datum

A Hecke algebra with a finite automorphic module or perfect complex, declared unramified Hecke polynomials, and a characteristic zero locus where classicality and an attached Galois representation are supplied. These realization inputs are part of the datum.

Chapter XII, Definition 9.1

See also: Hecke packet, Residual automorphic occupancy

B

Block diagonal plus low rank operator

An operator represented as a block diagonal part plus one thin factor times the adjoint of another. This separates independent packet work from a coupling term of rank at most the factor width. The abbreviation is BDLR.

Chapter VIII, Definition 3.3

See also: Packet operator representation, Output contract

Bridge datum

In the restricted Galois to type comparison, a supplied symplectic identification of two envelopes, compatible stabilizer action, and a common central character. At the prime two, compatible quadratic refinements and a lift to the central extensions are also required.

Chapter XI, Definition 11.1

See also: Symplectic envelope, Quadratic refinement

C

Clifford cocycle class

The cohomology class of the scalar factors by which multiplication of projective multiplicity operators differs from the group law. In the one break setting, its commutator pairing determines the associated symplectic envelope.

Chapter XI, Definition 9.3

See also: Symplectic envelope, Bridge datum

Common arithmetic spine

Algebraic Frobenius characteristic polynomials, coefficient-prime independent weight and determinant data, and prescribed ramified local labels that companion objects must share. Specifying the spine imposes compatibility conditions. It does not ensure that the corresponding residual fibers are occupied.

Chapter X, Definition 8.1

See also: Residual root, Residual forest, Automorphic Hecke datum

Compatibility

Agreement of specified constructions along two routes. For fine and coarse nonlinear models, applying the fine map and then coarsening must equal coarsening first and applying the coarse map. Arithmetic compatibility uses the separately declared common data.

Chapter III, Definition 20.1

See also: Refinement defect, Common arithmetic spine

Complete multiplicative filtration

A decreasing filtration by two-sided ideals whose products add filtration depths, whose intersection is zero, and whose compatible quotient system recovers the algebra. Positive transfer weights then control convergent expansions in the filtration topology.

Chapter II, Definition 5.1

See also: Weighted transfer series, Lift tree

Complete packet pruning

A set of nodes in a finite refinement tree meeting every root-to-terminal path exactly once, with no selected node above another. Parent projectors split into child projectors, so the selected packets still sum to the identity.

Chapter V, Definition 11.1

See also: Refinement, Packet system

Conditional expectation projector

For a fixed finite probability space and partition, the operator replacing a function by its probability-weighted mean on each partition block. It is an orthogonal projector in the corresponding weighted Hilbert space. Nested partitions yield compatible projections.

Chapter VI, Section 6

See also: Projector, Packet system

Congruential Euler coefficients

Coefficients defined by the reciprocal of an exponential section with width N and address j. The original degree-indexed coefficients and normalized packet coefficients differ by the factor j!. Both the normalization and the distinction between degree and row matter in congruences.

Chapter IX, Definition 3.1

See also: Packet row, Hurwitz integral series, p-adic precision

Connection

A chosen family of invertible maps between chart coordinates. The minimal connection is normalized at each chart. The stronger reversible connection also inverts transport on reversed edges. Neither definition alone imposes the cocycle law or identifies the connection with canonical coordinate transport.

Chapter III, Definition 5.1

Chapter III, Definition 5.5

See also: Flatness of a connection, Gauge change, Holonomy

Cyclic vector

A single vector whose orbit under the declared unital algebra generates the whole module by finite linear combinations. A nonzero module has seed rank one precisely when it has such a vector. Cyclicity is always relative to the acting algebra.

Chapter I, Proposition 4.2

See also: Seed, Seed rank, Generated hull

D

Derived support

For a perfect complex over a commutative ring, the prime ideals at which the derived tensor product with the residue field is nonzero. This support detects the complex after specialization, rather than relying on one selected cohomology module.

Chapter XII, Definition 6.14

See also: Module support, Ordinary patching datum

E

Exact lumpability

Compatibility between a fine Markov transition and a coarse transition under a declared aggregation map. The one-step intertwining identity makes fine evolution followed by aggregation agree with coarse evolution at every nonnegative integer time.

Chapter VI, Theorem 5.1

See also: Compatibility, Refinement defect, Transfer semigroup

F

Feshbach reduction

Elimination of hidden packets through their resolvent, producing an effective operator on the retained space. The correction records excursions through hidden packets. The exact resolvent identity requires the stated inverses and generally differs from bare compression.

Chapter VII, Theorem I.9.1

See also: Transfer block, Memory kernel

Flatness of a connection

The cocycle condition: transport through any intermediate chart equals the declared direct transport. In the reversible pair setting, flatness is equivalent to trivial triangle holonomy. Flatness alone does not say that the connection is already canonically calibrated.

Chapter III, Definition 5.6

See also: Connection, Holonomy, Gauge change

Forward difference

The difference between a sequence value after a declared step and its original value. Iterating this operation gives higher differences. In the Euler packet results the step acts on packet rows, so ordinary-degree displacement must be converted using the packet width.

Chapter IX, Definition 3.4

See also: Packet row, Congruential Euler coefficients, p-adic precision

G

Gauge change

A change of coordinate representative at each chart by an automorphism. It acts on a connection by conjugating the source and target representatives separately. Cocycle defects transform covariantly, while loop holonomy is conjugated at its base.

Chapter III, Definition 6.1

See also: Connection, Flatness of a connection, Holonomy

Generated hull

All finite sums of seed vectors acted on by elements of the declared algebra. In a topological realization, the topological hull is its closure. Algebraic generation and dense generation are distinct because closure may add limits with no finite addressed expression.

Chapter I, Definition 2.7

See also: Seed, Seed rank, Realization

Global deformation datum

A residual representation, local deformation rings and represented local-condition quotients, a global deformation ring, and compatible maps imposing those local conditions globally. Existence and representability of these objects require their own hypotheses. The datum does not establish automorphic occupancy.

Chapter XII, Definition 2.1

See also: Residual root, Residual automorphic occupancy, Ordinary patching datum

H

Hecke packet

The part of a finite Hecke module selected by a central idempotent of its finite-dimensional commutative Hecke algebra. It is nonzero precisely at an occupied maximal ideal. Over an algebraically closed field it is a simultaneous generalized eigenspace, retaining nilpotent data.

Chapter XII, Theorem 8.2

See also: Module support, Automorphic Hecke datum, Stable core

Holonomy

The composite transport around a loop for a chosen connection, based at its starting chart. Triangle holonomy measures the failure of a reversible pair connection to satisfy its cocycle law. It is not an invariant of an unspecified change of coordinates.

Chapter III, Definition 5.9

See also: Connection, Flatness of a connection, Gauge change

Hurwitz integral series

An exponential generating series whose factorial-basis coefficients are p-adic integers. Products and differentiation preserve this lattice. A series in it with unit constant coefficient has its reciprocal in the same lattice, supporting the reciprocal-transfer congruence arguments.

Chapter IX, Definition 4.1

See also: Congruential Euler coefficients, p-adic precision

K

Kraus channel

A finite sum of maps sending a matrix to K times that matrix times the adjoint of K, with the adjoint products summing to the identity. This normalization makes the resulting completely positive map trace preserving.

Chapter VII, Classical Input I.7.2

See also: Quantum seed system, Packet system

Krylov space

The span of a vector and its successive images under one operator, through a specified finite order. This records how repeated application enlarges the generated space. Packet decompositions interact with it under the invariance hypotheses stated in the relevant result.

Chapter VIII, Definition 7.1

See also: Cyclic vector, Generated hull, Packet operator representation

L

Lift tree

A rooted tree of approximation classes at increasing filtration depths. A child reduces to its parent and satisfies the next depth condition. Nonempty levels and finite branching provide a compatible branch. Completeness, separatedness and continuity are further needed to recover an exact solution.

Chapter III, Definition 23.5 and Theorem 23.6

See also: Complete multiplicative filtration, Obstruction class, Residual forest

M

Memory kernel

The time-dependent operator describing how eliminated packets return influence to retained packets. In the finite linear system it is a transfer into the hidden block, hidden evolution, and transfer back. Its Laplace transform is the corresponding self-energy under the stated spectral condition.

Chapter VI, Section 13.1 and Theorem 13.2

See also: Feshbach reduction, Transfer semigroup, Transfer block

Module support

The prime ideals of a commutative ring at which localization of the module is nonzero. Full support means every prime is occupied in this algebraic sense. It does not by itself provide a classical automorphic realization.

Chapter XII, Definition 6.1

See also: Near faithfulness, Derived support, Hecke packet

Monodromy packet chain

A finite sequence of packet spaces joined by a nilpotent operator that moves each packet down one level. A compatible Frobenius operator scales this transfer by the declared residue-field factor. Its powers are sums of paths along the chain.

Chapter XI, Definition 2.2 and Proposition 2.3

See also: Transfer block, Transfer graph

N

Near faithfulness

For a finite module over a Noetherian ring, nilpotence of its annihilator ideal. This is equivalent to full module support. If the ring is reduced it is also equivalent to faithfulness. The finiteness and ring hypotheses are part of the equivalence.

Chapter XII, Theorem 6.7

See also: Module support, Ordinary patching datum

O

Obstruction class

The class of a defect in the cokernel of a declared linear correction map. It vanishes exactly when that linear correction can remove the defect. The quotient, not the size of the defect, determines this solvability question.

Chapter III, Definition 21.1 and Theorem 21.2

See also: Refinement defect, Obstruction packet, Lift tree

Obstruction packet

A named packet with a certificate that the residual of one specified candidate is nonzero. It rules out that candidate under the declared solution condition. It neither constructs a solution nor proves that every possible candidate is obstructed.

Chapter III, Definition 7.1 and Theorem 7.2

See also: Obstruction class, Residual, Packet

Ordinary patching datum

A Noetherian patched ring, an ideal, and a finite patched module, together with identifications of the finite-level ring and module as their quotients. Full patched support descends under these hypotheses. Automorphy requires additional arithmetic data.

Chapter XII, Definition 6.10

See also: Module support, Near faithfulness, Global deformation datum

Output contract

The mathematical object an algorithm is required to return, such as an explicit vector, matrix, factors, or an implicit operator with an application rule. Costs are comparable only after the represented object and required output are fixed.

Chapter VIII, Definition 2.7

See also: Packet operator representation, Block diagonal plus low rank operator

P

p-adic precision

A lower bound on the p-adic valuation in the declared integral coefficient ring. The Euler congruence statements assign infinite valuation to zero. Divisibility in the field alone does not express precision, and special finite conventions for a zero step are stated separately.

Chapter IX, Definition 3.4

See also: Congruential Euler coefficients, Forward difference, Hurwitz integral series

Packet

In the linear theory, a component selected by a declared projector, usually identified with its range and index. Analytic, signal and arithmetic realizations attach their own additional structure. The meaning of a packet is therefore read with its specified system.

Chapter I, Definition 7.1

See also: Packet system, Projector, Transfer block

Packet analysis and synthesis

Analysis sends an object to its tuple of projected packet components. Synthesis adds those components in the ambient space. For a finite complete packet system the maps are inverse isomorphisms. Infinite versions require a specified convergence mode and corresponding bounds.

Chapter I, Definition 7.4 and Theorem 7.5

See also: Packet system, Packet support, Refinement

Packet jet

A higher derivative restricted to specified input packet spaces and projected to a specified output packet. Summing over all addresses reconstructs the derivative exactly. The analytic statements require the stated bounded projectors and continuous Fréchet differentiability.

Chapter III, Definition 17.1

See also: Packet, Transfer block

Packet operator representation

A finite packet layout, a declared set of active input-output addresses, and a linear block at each address. Evaluation extends each local block to the ambient direct sum and adds the blocks. Thus the representation has an explicit operator meaning.

Chapter VIII, Definition 2.2

See also: Transfer block, Output contract, Block diagonal plus low rank operator

Packet row

The index n within a fixed coefficient packet of width N. For the reciprocal Euler family, row n corresponds to ordinary degree Nn. A step of u rows therefore changes ordinary degree by Nu, a distinction needed when stating periods and difference estimates.

Chapter IX, Section 4, equations (4.1)-(4.2)

See also: Congruential Euler coefficients, Forward difference

Packet support

The set of packet indices on which an object has a nonzero projected component. A packet-pure vector has at most one such index, including the zero vector. This is different from module support in a prime spectrum.

Chapter I, Definition 7.2

See also: Packet, Seed rank, Module support

Packet system

A finite family of idempotent operators that annihilate one another and sum to the identity, giving an exact direct-sum decomposition into their ranges. Orthogonality is an additional condition when stated. Some chapters remove zero packet spaces by convention.

Chapter I, Definition 7.1

See also: Projector, Packet analysis and synthesis, Refinement

Perfect reconstruction

Recovery of every input by composing a filter bank with its synthesis bank, with any permitted gain and delay declared explicitly. The criterion is an identity of Laurent-polynomial matrices. A delay in decimated channel indices differs from a delay in original sample addresses.

Chapter V, Theorem 4.1

See also: Polyphase channel, Packet analysis and synthesis

Polyphase channel

The decimated sequence obtained by retaining one residue class of sample addresses and reindexing it on a coarse lattice. Its masked counterpart remains on the original lattice with other samples set to zero. Interleaving reconstructs the original sequence from all channels.

Chapter V, Section 2 and Theorem 2.1

See also: Residue selector, Perfect reconstruction

Projector

An idempotent linear operator: applying it twice has the same effect as applying it once. Its range is the selected component. The general packet calculus does not require an orthogonal projector, although Hilbert space and quantum results may impose self-adjointness.

Chapter I, Definition 7.1

See also: Packet system, Packet

Q

Quadratic refinement

Over the field with two elements, a quadratic function whose polar difference recovers a specified symplectic form. The alternating form alone does not determine the refinement. This additional datum controls dyadic phase information in the arithmetic packet constructions.

Chapter XI, Definition 13.1

See also: Symplectic envelope, Bridge datum

Quantum seed system

A finite Hilbert space with a nonzero seed, a declared operator grammar, constraints, packet systems at refinement levels, and the relevant dynamical and topological data. Its generated span is algebraic. Reachability by a restricted physical gate set is a separate question.

Chapter VII, Definition I.1.1

See also: Seed, Cyclic vector, Kraus channel

Realization

An assignment of spaces and linear maps to a typed operator grammar that satisfies its relations. More generally, a domain realization interprets specified abstract seed constructions in a chosen mathematical setting. Which operations and universal properties survive must be declared.

Chapter I, Definition 2.1

See also: Typed operator grammar, Realization functor, Seed

Realization functor

A functor from abstract seed systems to a declared domain category, equipped with a preservation profile. The profile states which identities or universal properties it preserves, such as sums, compositions, packet idempotents, quotients, or completions.

Chapter I, Definition 11.1

See also: Realization, Typed operator grammar

Reduced denominator

The one-variable entire function obtained from the signed exponential denominator by using a coordinate w equal to the Nth power of z. A root in the reduced coordinate records a root-of-unity orbit in the original variable, with multiplicities retained.

Chapter IV, Definition 5.7

See also: Residue selector, Transfer tangent, Zero power sum

Refinement

A finer packet system whose projectors sum to each coarse projector according to a surjective parent map. Fine packet ranges lie in their parent ranges. Exact refinement regroups complete component vectors and need not discard information.

Chapter II, Definition 3.2

See also: Packet system, Packet analysis and synthesis, Complete packet pruning

Refinement defect

The difference between applying a fine model and then coarsening, and coarsening first and applying the coarse model. Its vanishing expresses exact compatibility. A correction map must be specified before this vector defines a corresponding obstruction class.

Chapter III, Definition 20.1

See also: Compatibility, Obstruction class, Exact lumpability

Residual

In the candidate-certificate language, the value of a declared test at a candidate and packet. A solution makes every such value zero. This use of residual as a test value is distinct from reduction to a residue field in arithmetic deformation problems.

Chapter III, Section 7.1 and Definition 7.1

See also: Obstruction packet, Residual root

Residual automorphic occupancy

Existence of a nonzero localized automorphic or cohomological Hecke carrier realizing a specified residual representation and local conditions. It is an input at residual depth. Positive-depth deformations over another residual root do not supply it.

Chapter XII, Definition 2.4

See also: Residual root, Automorphic Hecke datum, Global deformation datum

Residual forest

The family of formal deformation problems above the admissible residual roots. Higher-depth objects are related by reduction, so each root has its own lifting branches. Declaring the forest does not establish that every intended branch or characteristic zero realization exists.

Chapter X, Definition 1.1

See also: Residual root, Lift tree, Common arithmetic spine

Residual root

An object over the residue field, fixed as the reduction of all objects in its residual fiber. Positive-depth corrections preserve this root. A universal arithmetic seed system therefore records all admissible roots rather than assuming one root represents every residual class.

Chapter X, Definition 1.1

See also: Residual forest, Residual automorphic occupancy, Visibility weight

Residue selector

The coefficient projector retaining exactly the exponents in one congruence class modulo a positive integer. It is defined directly by a coefficient mask and needs no division. A realization as a roots-of-unity average requires additional orthogonality hypotheses.

Chapter II, Definition 4.1

See also: Roots of unity filter, Polyphase channel, Refinement

Roots of unity filter

A finite character average realizing a residue selector when the required Fourier orthogonality certificate holds. Over a field with a primitive root, the usual normalization divides by the width. Over more general coefficient rings, the coefficient selector remains meaningful even when that average is unavailable.

Chapter II, Theorem 4.3

See also: Residue selector, Separation weight

S

Seed

A distinguished starting element acted on by a declared operator algebra or grammar. A seed family generates its hull through addressed operator expressions and finite sums, with closure added only in a specified topological regime. The acting algebra is fixed independently of the target.

Chapter I, Definition 2.6

See also: Generated hull, Seed rank, Typed operator grammar

Seed rank

The least size of a family generating a module under the declared acting algebra. In the topological version, generation is replaced by dense generation. The zero module has rank zero. Changing the acting algebra can change the seed rank of the same underlying space.

Chapter I, Definition 4.1

See also: Seed, Generated hull, Cyclic vector

Seed zeta function

The formal power series whose coefficient at each positive depth counts the finite objects above one residual root at that depth. Smooth and weighted charts yield explicit formulas under their stated assumptions. It records branching counts, not just the existence of a lift.

Chapter X, Definition 6.1

See also: Residual forest, Visibility weight

Separation weight

The least exponent of a uniformizer that makes a rational projector preserve a specified integral lattice, provided such an exponent exists. It records the precision needed to clear that projector on that lattice and is distinct from the visibility depth of two coordinates.

Chapter II, Definition 7.6

See also: Projector, Visibility weight, Roots of unity filter

Singular packet

In the signal chapter, a step or truncated-power component recording a value jump or a derivative jump at a declared breakpoint. Removing the trace jumps leaves the regular remainder covered by the stated Sobolev assumptions. Representation alone does not solve data-driven recovery.

Chapter V, Section E7

See also: Packet, Perfect reconstruction

Stable core

At one level of an inverse system of finite-dimensional spaces, the intersection of the images of every later level. It retains vectors that remain liftable arbitrarily far. Nonzero stable cores at every level provide compatible nonzero inverse-limit vectors, unlike mere nonvanishing of unrelated levels.

Chapter XII, Definition 8.9 and Theorem 8.11

See also: Hecke packet, Lift tree

Symplectic envelope

For a one break Clifford datum, the quotient of the finite elementary abelian group by the radical of its intrinsic commutator pairing. This removes the degenerate directions of that pairing. Comparison with a type-side envelope requires the additional bridge data.

Chapter XI, Definition 9.5

See also: Clifford cocycle class, Bridge datum, Quadratic refinement

Transfer block

The part of an operator carrying one input packet to one output packet: first select the input, apply the operator, and select the output. Summing all blocks reconstructs the operator for a finite complete packet system.

Chapter I, Definition 8.1

See also: Packet system, Transfer graph, Packet operator representation

Transfer graph

A directed graph on packet indices, with an edge where a declared algebra generator has a nonzero transfer block. Reachability limits where generated operators may have nonzero blocks. An allowed path does not prevent cancellation or guarantee a nonzero transfer.

Chapter I, Definition 9.1

See also: Transfer block, Seed rank, Packet operator representation

Transfer semigroup

A family of evolution operators whose composition adds time and whose value at zero is the identity. In the finite setting it is generated by a matrix exponential. The sign convention differs between generators of evolution and nonnegative diffusion operators.

Chapter VI, Definition 1.1 and Theorem 1.2

See also: Memory kernel, Exact lumpability

Transfer tangent

The negative logarithmic derivative of the signed exponential denominator in the cyclic analytic family. It is also the logarithmic derivative of the first reciprocal power. Its expansion connects reciprocal coefficient recurrences to logarithmic coefficients and reduced zero power sums.

Chapter IV, Definition 8.1

See also: Reduced denominator, Zero power sum

Typed operator grammar

A quiver of permitted source and target types together with linear relations among paths having the same endpoints. Its quotient path algebra encodes composition and relations. A realization assigns spaces to vertices and maps to arrows satisfying those relations.

Chapter I, Definition 2.1

See also: Realization, Seed, Transfer graph

V

Visibility weight

The first precision depth at which two distinct integral packet coordinates have different reductions, when such a depth exists. At the prime two, signs may share a residual reduction yet separate at a later depth. This weight measures separation of coordinates, not a projector denominator.

Chapter X, Definition 5.1

See also: Separation weight, Residual root, Seed zeta function

W

Weighted transfer series

A generating series for walks in a finite directed multigraph with positive integer edge weights and scalar multiplicities. It sums multiplicity products and endpoint weights at each total weight, including the empty walk. The positive weights make each coefficient well defined.

Chapter II, Definition 6.1

See also: Transfer graph, Complete multiplicative filtration

Z

Zero power sum

A sum of inverse powers of the reduced zeros of the cyclic analytic denominator, counted with multiplicity. For section order N at least two, these sums converge absolutely and agree with the normalized logarithmic coefficients through the genus-zero product.

Chapter IV, Definition 12.5 and Theorem 12.6

See also: Reduced denominator, Transfer tangent